

LIBRARY MANAGEMENT SYSTEM

ABSTRACT

The Library Management System is a simple C programming project developed to manage library operations digitally. This system helps in adding and displaying books easily. It reduces manual work and improves efficiency in maintaining library records. The project is user-friendly and useful for educational institutions and small libraries.

INTRODUCTION

Library Management System is developed using the C programming language to simplify the management of library records. In traditional libraries, maintaining records manually is difficult and time-consuming. This project provides an easy way to store and manage book information digitally. The system performs operations like adding books and displaying books.

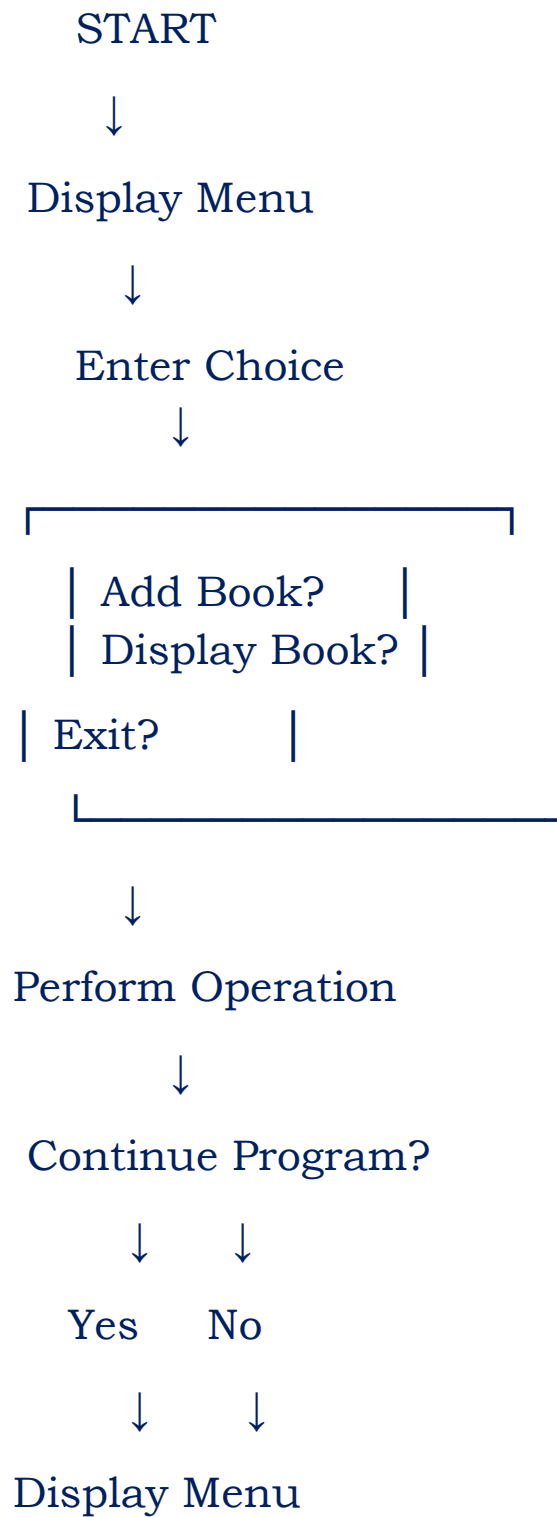
ALGORITHM

1. Start the program
2. Display menu options
3. Read user choice
4. Add or display books
5. Repeat until user exits

6. Stop the program



FLOWCHART



STOP

■ METHODOLOGY

The project is developed using the C programming language. The system stores book details such as book ID, book name, and author name. Different functions are used for adding and displaying books. The menu-driven approach makes the system easy to use and understand.

■ RESULT

The first screenshot shows the initial C code for a Library Management System. The code defines a struct for books, initializes an array, and implements an addBook function.

```
main.c
1 #include <stdio.h>
2 #include <string.h>
3
4 struct Book {
5     int id;
6     char name[50];
7     char author[50];
8 };
9
10 struct Book b[100];
11 int count = 0;
12
13 void addBook() {
14
15     printf("Enter Book ID: ");
16     scanf("%d", &b[count].id);
17
18     printf("Enter Book Name: ");
19     scanf("%s", b[count].name);
20
21     printf("Enter Author Name: ");
22     scanf("%s", b[count].author);
23
24     count++;
25 }
```

The second screenshot shows the code after adding a switch statement to handle user choices. The output shows the system running and successfully adding a book.

```
main.c
57
58     printf("Enter Your Choice: ");
59     scanf("%d", &choice);
60
61     switch(choice) {
62
63         case 1:
64             addBook();
65             break;
66
67         case 2:
68             displayBooks();
69             break;
70
71         case 3:
72             return 0;
73
74         default:
75             printf("Invalid Choice!\n");
76     }
77 }
78
79 return 0;
80 }
```

Output:

```
==== Library Management System ====
1. Add Book
2. Display Books
3. Exit
Enter Your Choice: |

==== Library Management System ====
1. Add Book
2. Display Books
3. Exit
Enter Your Choice: 1
Enter Book ID: 101
Enter Book Name: c programming
Enter Author Name: dennis ritchie

Book Added Successfully!

==== Library Management System ====
1. Add Book
2. Display Books
3. Exit
Enter Your Choice: 2

Book ID: 101
Book Name: c programming
Author Name: dennis ritchie

==== Library Management System ====
1. Add Book
```

■ **CONCLUSION**

The Library Management System successfully manages library records in a simple and efficient manner. It reduces manual work and helps users maintain book details properly. The project demonstrates the practical use of C programming concepts in real-world applications.

■ **FUTURE WORK**

In the future, this project can be improved by adding database connectivity, login authentication, fine calculation, and graphical user interface features. The system can also be converted into a web or mobile application.